

CROSS-MODAL DISTILLATION WITH MULTI-MODAL FUSION NETWORKS FOR REMOTE SENSING TARGET RECOGNITION UNDER MISSING MODALITIES

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ABSTRACT

Multimodal learning has set off a research hotspot in the remote sensing field due to its powerful performance. Although the paired multimodal training data can be collected offline, some of the data may be missing due to the limitation of the environment or the equipment in practical applications. Therefore, it is of great significance to effectively utilize the complete modal information in the training phase to assist the inference under missing modalities in the testing phase. To address this challenge, we propose a feature fusion-guided cross-modal distillation network. In our method, we design the modal interaction attention fusion module to mine the complementary information from multimodal data, as a multimodal learning model in the complete data scenario. When missing modalities, we use a combined knowledge distillation method to perform cross-modal learning. This is achieved by transferring sample feature similarity relationships and aligning logits output distributions, thereby reducing the redundancy of cross-modal learning feature representations. Finally, we use decision fusion to combine modality-specific information to complete the model inference. The proposed method is experimentally validated on the UNICORN dataset, which can effectively improve the model recognition rate under missing modalities.

Index Terms— remote sensing, multimodal learning, missing modalities, knowledge distillation, feature fusion.

1. INTRODUCTION

In recent years, multimodal learning has achieved great success in many visual tasks [1]. However, not all modalities are available in practical applications (testing phase), due to sensor failures, environmental interference, and data transmission issues [2]. Therefore, improving the target recognition performance of multimodal remote sensing images with missing modalities is of great practical significance.

At present, there are two main methods for the missing

modalities [3]: data imputation and hallucination model. The method based on data imputation operates at the data level, filling missing modal samples with random values, other similar modalities, or generating samples [4]. Modal generation methods refer to the use of generative models, such as auto-encoders, generative adversarial networks (GANs) to generate missing samples. The method based on data imputation is prone to introducing noise, and the sample quality generated by the model often falls short of the ideal state, requiring high storage and computation requirements.

The method based on hallucination model is currently a commonly used approach, aiming to generate missing modalities feature representations from available data. In 2016, J. Hoffman et al. [5] first proposed an hallucination model framework that significantly enhanced the accuracy of target recognition under missing modalities. This is achieved by simulating missing modal feature representations through feature and logits distillation. In 2020, Garcia et al [6] used generative adversarial networks to generate missing modal feature representations, improving the problem of introducing unnecessary noise from data generation.

The previously proposed hallucination models lack the extraction and utilization of multimodal complementary information, and the number of hallucination model branches in such methods increases with the number of missing modalities. In 2023, Wei et al. [7] proposed a general framework named MSH-net, which explores shared characteristics of modalities and solves the issues of missing modalities by reconstructing complete modal features from incomplete inference modalities.

Building on the aforementioned work, we propose a cross-modal knowledge learning method based on modal attention interaction fusion. The main contributions of this work are as follows:

(1) We propose a **cross-modal distillation with multi-modal fusion networks(CDMF-net)**, which can simultaneously address both complete and incomplete modalities.

(2) We propose a **modal interaction attention fusion method** that can effectively enhance and fuse the advantageous information between different modalities.

(3) We propose a **cross-modal knowledge learning method** through joint optimization of sample feature representation similarity and output distribution alignment.

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Section 2 introduces the details of our proposed method. Section 3 presents some experiments on the UNICORN dataset. Finally, the fourth section summarizes the conclusion.

2. PROPOSED METHOD

The main framework of our proposed method is shown in Fig. 1, which consists of a modal interaction attention module, a cross-modal learning module, and a decision fusion module. The overall framework is divided into four stages. Firstly, each single modality target recognition model is trained to obtain the basic model. Secondly, a multimodal fusion learning model is trained to obtain a teacher model based on the modal interaction attention module. Thirdly, train student model to imitate teacher model and train them based on cross-modal knowledge learning methods to obtain hallucination models. Finally, the hallucination model is fused with the base model for decision fusion and inference is completed.

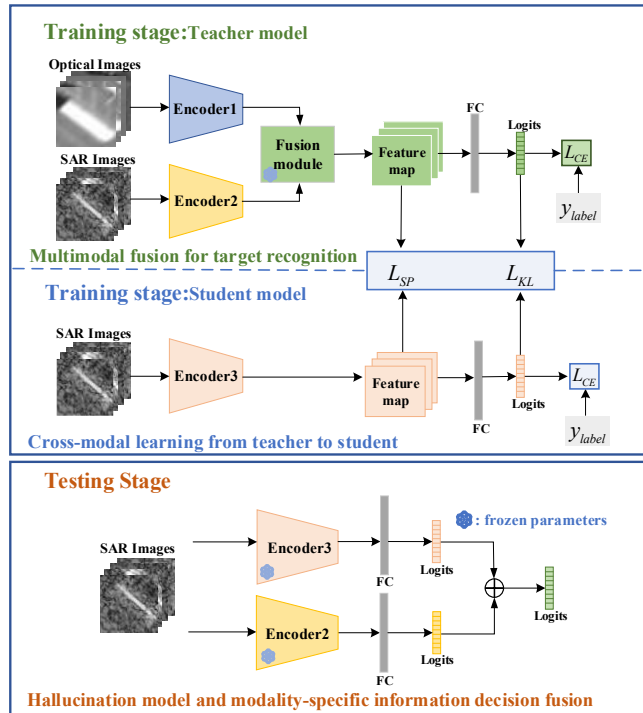


Fig. 1. The overall framework of proposed CDMF-net.

We will provide a detailed introduction to the three modules of our method in the following.

2.1. Modal interaction attention fusion module

Multimodal fusion can obtain a richer feature representation of the target and a more separable inter-class relationship. Taking inspiration from the effectiveness of cross attention mechanism in extracting information from different source

images, and integrating channel attention and spatial attention to augment image information, we propose a modal interactive attention fusion module, as shown in Fig. 2.

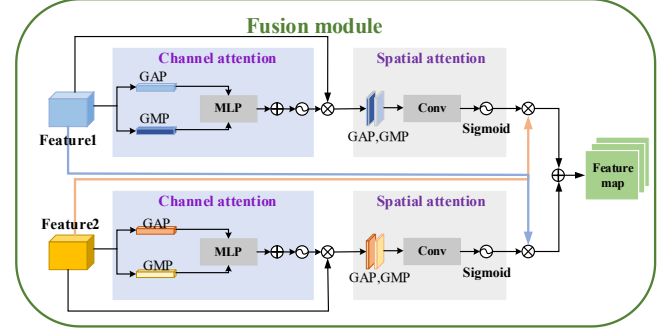


Fig. 2. The details of modal interaction attention fusion module.

There are existing the paired dataset containing modalities $X = \{X_1, X_2\}$. Let $\{f_1, f_2\}$ denote the feature maps of each modality. The feature maps obtain attention weights through channel attention respectively, which obtains channel enhanced feature map. The spatial information that different modality images focus on is different. In order to pay attention to this difference, the spatial weights are interactively multiplied into the modality feature map, and the interactive features of different modalities are added:

$$f'_i = \Phi_c(f_i) \otimes f_i \quad (1)$$

$$f_{fusion} = \Phi_s(f'_2) \otimes f_1 + \Phi_s(f'_1) \otimes f_2 \quad (2)$$

where i represents the i th modality, f'_i denotes enhanced feature maps, and $\otimes$ denotes element-wise multiplication. Here, f_{fusion} denotes the fusion output, Φ_c denotes channel attention, and Φ_s denotes spatial attention.

2.2. Cross-modal learning module

In the proposed framework, the hallucination model is used to learn fused features. Inspired by knowledge distillation[8], the fusion model is used as the teacher model. The hallucination model as a student model, since there is a large modal heterogeneity difference between the teacher model and the student model, it is difficult for feature-based knowledge distillation to ignore modal redundant information, resulting in overfitting learning. So we use the KL divergence to narrow the difference in the output distribution of the two models:

$$L_{KL}(p_{tea}, p_{stu}) = KL(p_{tea} \parallel p_{stu}) \quad (3)$$

$$KL(p_{tea} \parallel p_{stu}) = \frac{1}{N} \sum_{i=1}^n \frac{p_i^{tea}}{T} \log \left(\frac{p_i^{tea} / T}{p_i^{stu} / T} \right) \quad (4)$$

where T denotes the temperature, p_{tea} and p_{stu} represent the output probability distribution after the softmax function of teacher and student models.

We also align the two models based on their relationship of feature. Considering that there is a certain amount of redundant information between modalities, we used similarity to calculate the relationship between samples from the features, and then transferred the expression of the teacher's sample relationship to the student model. The theoretical basis of this method is that the similarity and dissimilarity relationships between samples are always consistent, and should also be consistent in both teacher and student models:

$$G_{tea} = \frac{F_t F_t^T}{\|F_t\| \|F_t^T\|}, G_{stu} = \frac{F_s F_s^T}{\|F_s\| \|F_s^T\|} \quad (5)$$

$$L_{SP} = MSE(G_{tea}, G_{stu}) \quad (6)$$

where F_t represents the feature maps of teacher model, F_s represents the feature maps of student model, and the KL divergence is used to narrow the difference in the output distribution of the two models.

Therefore, by combining the similarity relationship of samples and the output probability distribution, a comprehensive loss function expression is obtained:

$$Loss = \alpha(L_{CE}(p^{stu}, y_{label}) + L_{KL}(p^{tea}, p^{stu})) + \beta L_{SP}(f^{tea}, f^{stu}) \quad (7)$$

Where α and β denote parameters, and are used to keep the losses of each part at the same order of magnitude.

2.3. Decision Fusion inference module

After cross-modal learning, simulated fused feature expressions can be obtained. In order to better utilize existing modal information and retain the discriminative power of other modalities themselves, the output of the hallucination model is fused with specific modal information:

$$y_{fuse} = \arg \max(\text{soft max}((p_{hall} + p_i) / 2)) \quad (8)$$

when p_{hall} denotes the simulated feature fusion output probability distribution and p_i denotes the output probability distribution of the single modality.

3. EXPERIMENTAL RESULTS AND ANALYSIS

In this section, we will validate the performance of the proposed method through experiments conducted on the public dataset UNICORN [9]. This dataset, which has been manually labeled and aligned, was collected using airborne Electro Optical (EO) sensors and Synthetic Aperture Radar (SAR). The details of the data are listed in TABLE 1.

TABLE 1. THE TARGET INFORMATION OF UNICORN DATASET.

	0	1	2	3	4
Training	1000	1000	1000	1000	1000
Test	200	200	200	200	200
	5	6	7	8	9
Training	652	628	424	640	433
Test	200	200	200	200	200

In this section, we will compare the recognition results of our method with other methods and conduct ablation experiments.

3.1. Comparisons with other methods

We compare with relevant methods for missing modalities, including ADDA [6], MSH-net [7], MARS[10], DASK [11], MDMS-net [12]. To ensure fairness in the experiment, we used Resnet50 as the backbone and repeated the experiment three times to take the average result. The recognition results of 10 types of vehicle target are shown in TABLE 2.

TABLE 2. COMPARATIVE EXPERIMENTAL RESULTS OF DIFFERENT METHODS.

Methods	Training modalities	Test modalities	Accuracy
Baseline	EO	EO	91.68%
	SAR	SAR	80.75%
Fusion(ours)	EO,SAR	EO,SAR	93.47%
ADDA[6]	EO,SAR	SAR	83.97%
MSH-net[7]	EO,SAR	SAR	84.20%
MARS[10]	EO,SAR	SAR	81.33%
DASK[11]	EO,SAR	SAR	81.77%
MDMS-net[12]	EO,SAR	SAR	83.52%
CDMF-net(ours)	EO,SAR	SAR	85.37%

In TABLE 2, baseline is the recognition result of a single modality, and fusion experiment using two complete modalities, which is also the teacher model in the subsequent method. In addition, all experimental results when the optical is missing. Given that the recognition accuracy for the optical modality alone are significantly higher than that for the SAR modality alone, enhancing the performance of the SAR modality is particularly meaningful. The proposed method achieves a higher recognition rate than the comparative method.

3.2. Ablation study

TABLE 3 presents the ablation experimental results of each module. The experimental results show that the combination of relational and logits distillation can make transfer learning more sufficient; modality-specific information also has certain supplementary capabilities, which can help the hallucination model correct some wrong decisions. As a teacher model, the multi-modal fusion model has richer target information and effectively improves the recognition rate of the student model.

We perform visual analysis on the feature maps as shown in Fig. 3. The feature separability is further enhanced after the fusion model. After cross-modal learning, the hallucination model is better than the SAR-single, significantly improving intra-class aggregation.

TABLE 3. THE EXPERIMENTS OF ABLATION STUDY.

Fusion	Modules			Accuracy
	Logits distillation	Relation distillation	Decision fusion	
×	×	×	×	80.75%
✓	✓	×	×	82.72%
✓	✓	✓	×	83.50%
×	✓	✓	✓	85.00%
✓	✓	✓	✓	85.37%

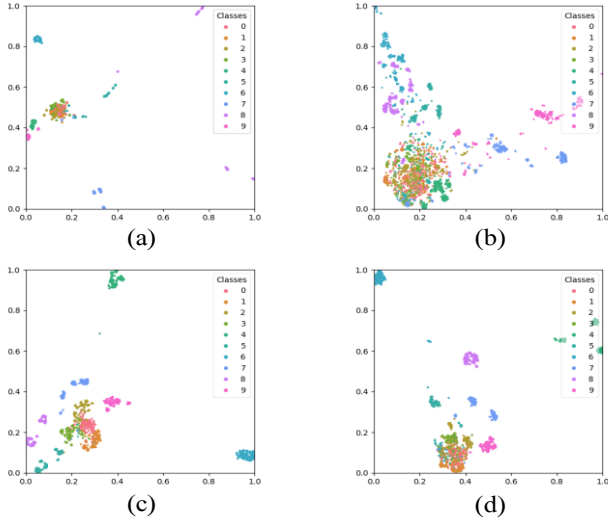


Fig. 3. t-SNE visualization of internal feature representation learned by various methods on the testing set: (a)EO-single, (b)SAR-single, (c)fusion model, (d) hallucination model.

To verify the effectiveness of our proposed fusion module, we conducted some comparative experiments with other fusion methods, such as additive fusion, channel concatenation, transformer fusion and channel interaction fusion. As shown in TABLE 4, the proposed fusion module achieves the best performance with complete modalities.

TABLE 4. THE EXPERIMENTS OF FUSION MODULE.

Fusion module	Accuracy
Add	92.40%
Concatenation	92.98%
Cross-channel attention	92.05%
Transformer	93.11%
Ours	93.47%

4. CONCLUSION

Our method extracts advantageous information between different modalities by enhancing modal interaction, and jointly optimizes learning from sample feature similarity relationships and logits output probabilities, effectively improving the feature extraction and category differentiation capabilities of single modality. The proposed network can simultaneously

improve the target recognition performance of complete modalities and incomplete modalities, thereby achieving robust multi-modal fusion recognition. Experiments on the UNICORN data set show that our method performs better than other methods and can effectively reduce the impact of missing modalities.

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